Basic Information

Paper Title / Working Title: AI-Powered Space Mission Risk Prediction / Flood Monitoring Framework
Conference / Track: — GLOC
Lead Author: — Karthikeya moyye
Team Members: — ankit thakur, abdullah, Mansi gupta
Week Number / Dates: Week 1 / (23/01-01/02)

Progress This Week

Writing Progress

Sections written/revised:
Worked on consolidating literature review inputs from team members and identifying core research gaps. Main focus was on structuring the problem space and aligning individual contributions into a unified research direction. (~500–600 words drafted in rough form)

Figures, tables, or diagrams prepared:
No figures prepared this week. Focus remained on conceptual and textual structuring before visual representation.

Research & Analysis

Literature reviewed/added:
Reviewed prior work on flood detection using SAR, optical imagery, and hybrid models. Key insights included limitations of single-sensor pipelines, computational bottlenecks in SAR preprocessing, and lack of real-time modular systems. These findings support the need for a multi-sensor, modular preprocessing framework.

Data collected/analyzed:
No new datasets analyzed this week. Focus remained on conceptual framework design and gap identification.

Conceptual Development

Major decisions made:
Shifted focus from single-model optimization to system-level architecture design (modular preprocessing + multi-sensor fusion + scalable deployment). This direction was chosen to improve real-world deployability and long-term scalability.

Methodology refined:
Refined the approach to emphasize hybrid SAR–optical fusion, lightweight preprocessing, and AI-assisted pattern recognition instead of heavy monolithic pipelines. This was driven by limitations observed in existing literature.

Current Status

On track:
The project direction and research gaps are now clearly defined. Conceptual clarity has improved, and the team has a shared understanding of the core problem and proposed direction for development.

Challenges & Blockers

Scientific/Technical Issues

Integration of multi-sensor data pipelines conceptually clear, but implementation-level design is still undefined.

Writing & Clarity Issues

Individual team inputs are fragmented and need harmonization into a single coherent narrative.

References & Data Gaps

Need more recent (post-2023) papers on hybrid SAR–optical fusion and real-time flood monitoring systems.

Team Coordination

Inputs are being generated individually rather than collaboratively structured, creating flow and consistency issues.

Support Requested

Conceptual guidance needed on:
Designing a scalable modular preprocessing architecture.

Literature suggestions for:
Hybrid SAR–optical fusion models and real-time flood monitoring systems.

Feedback requested on:
Literature review flow and gap articulation structure.

Technical review needed for:
Overall system architecture logic (preprocessing + AI fusion + deployment model).

Plan for Next Week

Writing Goals

Sections to complete:
Data sets required and possible ML models for flood prediction as assessed by metrics(~800–1000 words)

Revisions planned:
Improve logical flow and unify tone across contributions

Research Goals

Literature to review:
2-4 papers on SAR-optical fusion, modular pipelines, and real-time flood systems

Data/analysis tasks:
Conceptual design of preprocessing pipeline architecture

Deliverables

Internal review target:
- Draft literature + ML model discussion report

Figures/tables to finalize:
- Ml model metrics table for each prospective ML models that are actively used for flood detection

Quality Assurance Self-Check

Plagiarism Prevention

☑ All content paraphrased in own words
☑ No direct copying from sources

Scientific Accuracy

☑ Claims aligned with reviewed literature
☑ No unsupported conclusions

Citation Quality

☑ Primary research sources prioritized
Number of new references added this week: 10

Optional: Reflections

This week helped clarify the real research direction and exposed structural weaknesses in existing systems rather than model-level issues alone. Greater coordination and shared structuring will improve both writing quality and conceptual coherence going forward.

Submitted by: Karthik
Date: 23/02/2026